

Vector Stores

Complete Guide to Vector Storage in LangChain

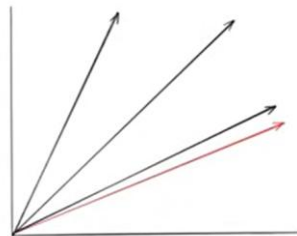
Why Vector Stores? - Movie Database Example

Movie Metadata Table

Movie ID	Movie Name	Director	Actor	Genre	Release Date
M001	3 Idiots	Raju Hirani	Aamir Khan	Drama, Romance	2009
M002	Chennai Express	Rohit Shetty	Shah Rukh Khan	Romance, Comedy	2014
M003	Inception	C Nolan	L DiCaprio	Thriller, Sci-Fi	2009
M004	Stree	Amar Kaushik	Rajkummar Rao	Horror, Comedy	2019

Vector Concept:

Each movie can be converted into a numerical vector using embeddings. Vector stores then organize these vectors in a multi-dimensional space, enabling fast similarity searches. For example, searching for movies similar to 'Inception' would find movies with similar genres, themes, and metadata.



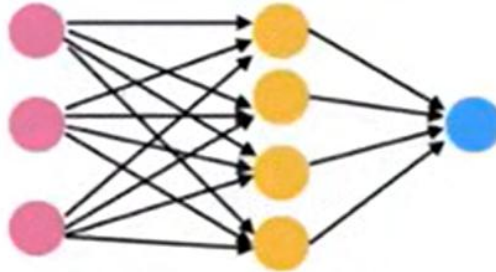
0.32	0.63	0.13	0.45	0.0	0.1	• • • • •	0.78	0
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0.44	0.61	0.0	0.91	0.0	0.99	• • • • •	0.91	0
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Movie id	Plot
M001	In the present day, Farhan receives a call from Chatur, saying that Ranchi is coming. Farhan is so excited that he fakes a heart attack to get off a flight and picks up Raju from his home (who forgets to wear his pants). Farhan and Raju meet Chatur at the water tower of their college ICE (Imperial College of Engineering), where Chatur informs them that he has found Rancho. Chatur taunts Farhan and Raju that now he has a mansion worth \$3.5 million in the US, with a heated swimming pool, a maple wood floor living room and a Lamborghini for a car. Chatur reveals that Rancho is in Shimla...
M002	Rahul Mithaiwala (Shahrukh Khan) is a forty-year old bachelor who lives in Mumbai. His parents died in a car accident when he was eight years old and was brought up by grandparents. His grandfather has a sweet-selling chain store - Y.Y. Mithaiwala. Before his birth centenary celebration, two of Rahul's friends suggest a vacation in Goa which he accepts. On the eve of the celebration, his grandfather dies whilst watching a cricket match. His grandmother tells him that his grandfather desired to have his ashes divided into two parts - to be immersed in the Ganges River and Rameswaram respectively. She requests Rahul to go to Rameswaram and immerse them. Rahul reluctantly accepts her request but was also eager to attend the Goa trip...

In the peculiar town of Chanderi, India, the residents believe in the myth of an angry woman ghost, referred to as "Stree" (Hindi for woman) (Flora Saini), who stalks men during Durga Puja festival. This is explained by the sudden disappearance of these men, leaving their clothes behind. She is said to stalk the men of the town, whispering their names and causing disappearances if they look back at her.

The whole town protects itself from Stree during the 4 nights of Durga Puja by writing "OO Stree, Kal Aana" on their walls. Additionally, men are advised to avoid going out alone after 10 PM during the festival and to move in groups for safety. This practice reflects a societal parallel to the precautions typically advised to women for their own protection....



0.33	0.03	0.10	0.22	0.99	0.77
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Challenges & Solution

- Generate Embedding Vectors
- Storage
- Semantic Search – Intelligent Semantic Search

- Solution
 - Vector Store

Vector Representation - Movie Plots as Embeddings

3 Idiots

Genre: Drama, Romance

A drama about three engineering students and their pursuit of knowledge and friendship.

Embedding Vector: [0.92, 0.83, 0.54, 0.88, 0.91, ...]

Inception

Genre: Thriller, Sci-Fi

A sci-fi thriller about corporate espionage and dream infiltration technology.

Embedding Vector: [0.87, 0.45, 0.92, 0.51, 0.88, ...]

Stree

Genre: Horror, Comedy

A horror-comedy about a ghostly woman who haunts men in a small town.

Embedding Vector: [0.76, 0.68, 0.62, 0.84, 0.55, ...]

What is a Vector Store?

A vector store is a system designed to store and retrieve data represented as numerical vectors.

Storage

Ensures that vectors and their associated metadata are retained, whether in memory for quick lookups or on-disk for durability and large-scale use.

Similarity Search

Helps retrieve the vectors most similar to a query vector

Indexing

Provide a data structure or method that enables fast similarity searches on high-dimensional vectors (e.g., approximate nearest neighbor lookups).

CRUD Operations

Manage the lifecycle of data—adding new vectors, reading them, updating existing entries, removing outdated vectors.

Vector Store Use Cases



Semantic Search

Find documents based on meaning, not just keywords



RAG (Retrieval-Augmented Generation)

Retrieve relevant context to improve LLM responses



Recommender Systems

Suggest similar products, content, or users



Image/Multimedia Search

Find similar images or multimedia based on embeddings

Vector Store vs Vector Database

Vector Store

- Storage & Retrievers, Lightweight library or service
- Focuses on storing embeddings & similarity search
- May not include many traditional database features like transactions, rich query languages, or role-based access control
- Ideal for prototyping
- Example: FAISS (where you store vectors and can query them by similarity, but you handle persistence and scaling separately)

Vector Database

- Full-fledged database system
- Advanced database features : nearly ACID, authentication / Authorization, Durability & Persistence, Distributed architecture
- Distributed & scalable
- Production-ready
- Examples: Milvus, Qdrant, Weaviate, Pinecone

A vector database is effectively a vector store with extra database features (clustering, scaling, security, metadata filtering, and durability)

Vector Stores in LangChain



Multiple Integrations

FAISS, Pinecone, Chroma, Qdrant, Weaviate, and many more



Common Interface

Uniform API allows swapping backends with minimal code changes



Metadata Handling

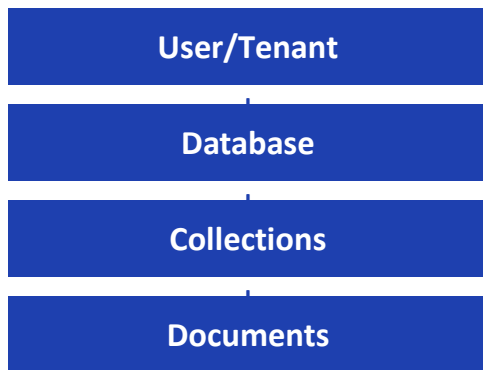
Attach metadata (timestamps, authors, tags) to each document

```
from_documents()  
add_documents()  
similarity_search(query,k=2)  
metadata-Based Filtering
```

Chroma Vector Store

Chroma is a lightweight, open-source vector database that is especially friendly for local development and small- to medium-scale production needs.

Chroma Hierarchy



Key Features:

- ✓ Open-source and lightweight
- ✓ Perfect for development environments
- ✓ Multi-tenant support with hierarchical organization

Tenant: User workspace
 Database: Isolated data store
 Collection: Related data group
 Doc: Individual record

Key Takeaways

1. Vector stores enable efficient storage and retrieval of vector embeddings from various data sources
2. Convert data (text, images, metadata) into vectors for semantic similarity searches
3. Vector stores are lightweight; vector databases offer enterprise-grade scalability
4. LangChain provides unified interfaces to multiple vector store backends
5. Choose based on your needs: Vector stores for development, databases for production